Collision-Compressed Prime-Shell Reassembly on Finite Windows

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Abstract

We prove a collision-compressed finite-window envelope for the exact TPC-218 common-source packet kernel. The previous scalar recovery collapsed the prime-shell label by a factor $P = \#\mathcal{Q}_x$. We instead combine the prime rows inside each primitive physical frequency bucket before applying the reduced-frequency additive large sieve. The TPC-236 incidence theorem gives the uniform bucket factor

$$R_* = \frac{4Q^2}{H} + \frac{4UQ}{H},$$

while the direct coefficient energy is $O(JM^2(Q^2/H)(\log x)^5)$. On $I_x = (x/2, x] \cap \mathbb{Z}$ this yields

$$\frac{1}{|I_x|} \sum_{n \in I_x} \sum_{j=1}^{J} |K_j(n)|^2 \ll JM^2\big(x^{1/48} + x^{1/50}\big)(\log x)^5.$$

The leading unnormalized exponent is $49/48 + o(1)$, and $U^2/|I_x| = x^{-67/200+o(1)}$ makes the finite-window correction lower order. An exact V59-shaped, source-active fixture at $h = 82$ independently reproduces the collision and window composition. The theorem controls an unsigned packet trace. It proves neither signed divisor cancellation nor the signed four-packet Gate-B scalar, and it makes no sharpness claim.

Claim level. `PROVED_STRUCTURAL_L1`. The certified object is the common-source, collision-compressed finite-window packet trace.

1 Introduction

The common-source finite-window kernel retains two outer labels that should not be discarded casually: the prime-shell row q and the packet profile j. TPC-218 kept both labels through a Hilbert-valued large-sieve estimate, but recovering the scalar prime-shell sum by pointwise Cauchy cost $P = \#\mathcal{Q}_x$ and returned the normalized power $x^{11/32}$ [3]. The resulting bound was source-valid but ignored the fact that only a small number of physical prime rows can meet one residue coordinate.

TPC-236 supplied the missing local geometry. At a fixed denominator h, a residue a belongs to at most

$$\frac{4Q^2}{H} + \frac{4hQ}{(a,h)H}$$

prime rows [4]. The finite-window kernel already uses reduced primitive frequencies. Thus $(a,h)=1$, and the collision theorem can be inserted exactly at the scalar-recovery step. This insertion is made before, rather than after, the additive large sieve.

The changed order replaces the coarse factor P by $R_* = 4Q^2/H + 4UQ/H$. Combining this factor with the inherited direct coefficient energy produces powers $x^{1/48}$ and $x^{1/50}$, a substantial structural reduction from $x^{11/32}$. The same primitive coordinates remain separated by U^{-2}, so the finite-window attachment of TPC-217 applies without changing the physical object [2].

The paper establishes three precise points:

1. a collision Bessel inequality for the literal q sum on primitive coordinates;
2. its composition with the reduced-frequency large sieve and the explicit C_h coefficient energy;
3. an exact exponent ledger and an independently checked source-active finite fixture.

Every step is unsigned. In particular, the theorem does not convert the packet trace into the signed four-packet Gate-B scalar and does not exploit signs of C_h.

2 Frozen common-source object

Set

$$H = x^{21/32}, \qquad Q = x^{1/3}, \qquad U = x^{133/400}, \tag{1}$$

and define

$$\mathcal{Q}_x = \{q \text{ prime} : Q < q \le 2Q\}, \tag{2}$$

$$\mathcal{D}_x = \{d : H/(4Q) < d \le U,\ \mu(d)^2 = 1\}, \tag{3}$$

$$C_h = \sum_{\substack{d\in\mathcal{D}_x \\ h\mid d}} \frac{\mu(d)\log d}{d}. \tag{4}$$

Let J be fixed. For bounded profiles ψ_j, write $M = \max_j \|\psi_j\|_\infty$ and

$$B^{(j)}_{h,q}(a) = \sum_{0<|m|\le\lfloor hq/H\rfloor} \psi_j\left(\frac{Hm}{hq}\right) \mathbf{1}_{mq^{-1}\equiv a \pmod h}. \tag{5}$$

The object studied here is

$$K_j(n) = \sum_{h\le U} \sum_{\substack{a \bmod h \\ (a,h)=1}} C_h \left(\sum_{q\in\mathcal{Q}_x} B^{(j)}_{h,q}(a)\right) \mathrm{e}^{2\pi i n a/h}. \tag{6}$$

Equation (6) freezes all interfaces used below. The outer q weight is one, C_h is the literal signed divisor coefficient, and no row-dependent normalization or packet-dependent transform is inserted. The pairs (h,a) are primitive at the point where they enter the large sieve. Passing all residues to that step would duplicate rational frequencies and is therefore forbidden.

For sufficiently large x, $U < Q$ and $4Q < H$. Hence every shell prime is invertible modulo every $h \le U$. The frequency $0/1$ contributes no atom because $\lfloor q/H\rfloor = 0$ for $q \le 2Q$.

3 Prime-shell collision compression

Let $R_h(a)$ count the shell rows for which $B_{h,q}^{(j)}(a)$ can be nonzero. TPC-236 proves its gcd-fiber bound uniformly in the row amplitudes. Restricting that incidence matrix to primitive coordinates gives the following consequence.

Lemma 3.1 (Primitive collision factor)**.** *For every frequency in* (6)*,*

$$R_h(a) \leq \frac{4Q^2}{H} + \frac{4hQ}{H} \leq \frac{4Q^2}{H} + \frac{4UQ}{H} =: R_*. \tag{7}$$

Consequently,

$$\sum_{h,a,j} |C_h|^2 \left| \sum_{q\in\mathcal{Q}_x} B_{h,q}^{(j)}(a) \right|^2 \leq R_* \sum_{h,a,j,q} |C_h B_{h,q}^{(j)}(a)|^2. \tag{8}$$

All a-sums in this paper are over $(a,h)=1$.

Proof. The physical multiplicity theorem has second term $4hQ/(gH)$ with $g=(a,h)$. Here $g=1$, which proves (7). At one fixed (h,a,j), at most $R_h(a)$ values of q contribute. Cauchy–Schwarz therefore bounds the squared q sum by $R_h(a)$ times the sum of the individual squares. Multiplication by $|C_h|^2$ and summation prove (8). □

The right side of (8) is already controlled by the source.

Lemma 3.2 (Direct coefficient energy)**.** *Under the scales* (1)*,*

$$\sum_{h,a,j,q} |C_h B_{h,q}^{(j)}(a)|^2 \ll JM^2 \frac{Q^2}{H} (\log x)^5. \tag{9}$$

Proof. The condition $4Q < H$ makes $m \mapsto mq^{-1} \pmod h$ injective on the range in (5). Thus

$$\sum_{(a,h)=1} |B_{h,q}^{(j)}(a)|^2 \leq 2M^2 \frac{hq}{H}.$$

If this row is active, then $h \geq H/(2Q)$. Writing $d = hk$ in (4) and discarding the lower band and squarefree restrictions gives

$$|C_h| \leq \frac{\log U}{h} \sum_{k \leq U/h} \frac{1}{k} \ll \frac{(\log x)^2}{h}.$$

It follows that

$$\sum_{h \text{ active}} h|C_h|^2 \ll (\log x)^4 \sum_{H/(2Q)\leq h \leq U} \frac{1}{h} \ll (\log x)^5.$$

Finally $q \leq 2Q$ and $\#\mathcal{Q}_x \leq 2Q$. Summing the row estimate over j,q,h proves (9). □

4 Finite-window theorem

The remaining analytic input is the additive large sieve in its standard separated- frequency form [1]. If the frequencies are separated modulo one by δ, then on any consecutive interval I of N integers,

$$\sum_{n\in I} \left| \sum_{\ell} z_\ell e^{2\pi i n \alpha_\ell} \right|^2 \leq (N - 1 + \delta^{-1}) \sum_{\ell} |z_\ell|^2. \tag{10}$$

Lemma 4.1 (Primitive spacing)**.** *Distinct primitive fractions a/h and a'/h' with $h, h' \le U$ have circular distance at least U^{-2}.*

Proof. Their difference, or its complement modulo one, is a nonzero rational whose denominator divides hh'. Its numerator is a nonzero integer, so the distance is at least $(hh')^{-1} \ge U^{-2}$. □

Theorem 4.2 (Collision-compressed finite-window reassembly)**.** *For every consecutive interval I of N integers,*

$$\sum_{n\in I}\sum_{j=1}^{J} |K_j(n)|^2 \le (N-1+U^2)R_* \sum_{h,a,j,q} |C_h B_{h,q}^{(j)}(a)|^2, \tag{11}$$

$$\ll (N+U^2)JM^2R_*\frac{Q^2}{H}(\log x)^5. \tag{12}$$

Proof. Apply (10) for each packet j to the coefficients

$$z_{h,a}^{(j)} = C_h \sum_{q\in\mathcal{Q}_x} B_{h,q}^{(j)}(a).$$

Lemma 4.1 permits $\delta^{-1} \le U^2$. Summing in j and applying Lemma 3.1 proves (11). Lemma 3.2 then gives (12). □

The order of the proof is load-bearing. Collapsing q by the unconditional bound $|\sum_q z_q|^2 \le P\sum_q |z_q|^2$ would restore the TPC-218 exponent. Here physical occupancy replaces P before the finite-window Gram is diagonalized.

5 V59 exponent ledger and method boundary

Take $I = I_x = (x/2, x] \cap \mathbb{Z}$ and $N = |I_x|$. The exact exponent identities are

$$\frac{Q^2}{H} = x^{1/96}, \qquad \frac{UQ}{H} = x^{23/2400}, \tag{13}$$

$$\left(\frac{Q^2}{H}\right)^2 = x^{1/48}, \qquad \frac{UQ}{H}\frac{Q^2}{H} = x^{1/50}, \tag{14}$$

$$\frac{U^2}{N} = x^{-67/200+o(1)}. \tag{15}$$

Dividing (12) by N therefore yields

$$\frac{1}{N}\sum_{n\in I_x}\sum_j |K_j(n)|^2 \ll JM^2\big(x^{1/48}+x^{1/50}\big)(\log x)^5 \ll JM^2x^{1/48}(\log x)^5. \tag{16}$$

The leading unnormalized exponent is $1 + 1/48 = 49/48$.

Interface	Normalized power	Information retained
TPC-218 split packet array	1/96	separate q, j labels
TPC-218 scalar recovery	11/32	unsigned scalar trace
Theorem 4.2	1/48 (main)	primitive physical occupancy

The comparison records a structural upper bound, not a sharp asymptotic. The proof sequentially composes a collision Bessel inequality, an absolute C_h harmonic majorant, and the additive large sieve. Valid upper bounds need not be simultaneously saturated by one physical source. Conversely, no step uses the signs in C_h.

The terminal firewall is therefore strict:

$$\text{unsigned packet trace} \neq \text{signed four-packet Gate-B scalar.}$$

No arithmetic L^2, fixed-atom credit, strict 1/400 payment, full Gate B, or twin-prime conclusion follows from (16).

6 Independent finite reproduction

The certificate uses

$$(Q, H, U, h) = (101, 8830, 99, 82).$$

Exact integer-power comparisons give

$$8830^{32} \leq 101^{63} < 8831^{32}, \qquad 99^{400} \leq 101^{399} < 100^{400}.$$

Thus the fixture reproduces the V59 floor exponents. Unlike the earlier $h = 80$ collision adversary, $h = 82 = 2 \cdot 41$ is squarefree and source-active. In the band $H/(4Q) < d \leq U$, its rational marked-divisor reproduction has the single term

$$C_{82}^{\mathrm{rat}} = \frac{\mu(82)}{82} = \frac{1}{82}.$$

The replacement of $\log d$ by 1 makes the finite ledger rational; it is not used in the asymptotic theorem.

For $q = 109, 137, 191$, every row has primitive support $\{3, 79\}$. With the constant packet and the signed-multiplier packet, the direct energy and collapsed trace are

$$E_{\mathrm{dir}} = \frac{3}{1681}, \qquad E_{\mathrm{col}} = \frac{5}{1681}, \qquad \frac{E_{\mathrm{col}}}{E_{\mathrm{dir}}} = \frac{5}{3}.$$

The constant packet alone has collision ratio three. On the complete 82-point window, orthogonality of 3/82 and 79/82 gives exact trace energy 10/41, hence normalized energy 5/1681.

The producer and a separately implemented checker reconstruct the floor relations, Möbius band, modular rows, primitive frequencies, energies, collision factor, and large-sieve composition. A third program tests 21 shifted intervals, including lengths 82 and 164. Normal and optimized Python runs have identical output. The certificate record digest is

`41acb31d49f21e305e97941038b96ef0f6c938474e1e02cb0085f9ae01044848`.

Finite agreement certifies the implementation and displayed algebra only.

7 Conclusion

Physical collision compression can be performed before reduced-frequency finite-window attachment on the exact TPC-218 common-source kernel. Primitive coordinates turn the TPC-236 gcd-fiber estimate into the factor $4Q^2/H + 4UQ/H$. After the direct coefficient-energy bound and additive large sieve, the normalized packet-trace exponent becomes 1/48, with a secondary 1/50 term and a lower-order window correction.

The strongest remaining obstruction is now narrower. Estimate (8) uses only a uniform occupancy maximum, while Lemma 3.2 discards all signs through $|C_h|^2$. The next admissible theorem should test the literal weighted collision energy

$$\sum_{h,j} \sum_{(a,h)=1} |C_h|^2 \left| \sum_q B_{h,q}^{(j)}(a) \right|^2$$

against the product of R_* and the direct energy. A strict saving would identify new source structure; a matching source-valid fixture would establish an obstruction. Signed cross-denominator cancellation should be attempted only after this test.

Reproducibility and declarations

Data and code availability. All mathematical source files, exact certificate data, and deterministic checking programs are included in the TPC-237 project directory. No external dataset is used.

Author contributions. Liang Wang: conceptualization, methodology, formal analysis, software, validation, and writing.

Ethics. This theoretical and computational study involves no human participants, animals, or sensitive personal data.

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